

Diarization-Aware Multi-Speaker Automatic Speech Recognition via Large Language Models

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Abstract—Multi-speaker automatic speech recognition (MS-ASR) faces significant challenges in transcribing overlapped speech, a task critical for applications like meeting transcription and conversational analysis. While serialized output training (SOT)-style methods serve as common solutions, they often discard absolute timing information, limiting their utility in time-sensitive scenarios. Leveraging recent advances in large language models (LLMs) for conversational audio processing, we propose a novel diarization-aware multi-speaker ASR system that integrates speaker diarization with LLM-based transcription. Our framework processes structured diarization inputs alongside frame-level speaker and semantic embeddings, enabling the LLM to generate segment-level transcriptions. Experiments demonstrate that the system achieves robust performance in multilingual dyadic conversations and excels in complex, high-overlap multi-speaker meeting scenarios. This work highlights the potential of LLMs as unified back-ends for joint speaker-aware segmentation and transcription.

Index Terms—Multi-Speaker Automatic Speech Recognition

I. INTRODUCTION

Automatic Speech Recognition (ASR) [1]–[3] for single-speaker scenarios has achieved remarkable success and widespread deployment in industrial applications. However, real-world conversations, such as meetings, interviews, and discussions, often involve multiple speakers. In such settings, conventional ASR systems fail to distinguish who spoke what, rendering them insufficient for tasks requiring speaker-specific content understanding, such as meeting summarization and speaker-centric analysis.

Multi-Speaker ASR (MS-ASR) [4]–[9] extends the conventional ASR task by not only transcribing speech content but also attributing each utterance to the correct speaker. Unlike standard ASR, which assumes a single active speaker, MS-ASR must operate in conversational scenarios where multiple speakers take turns or speak simultaneously. The core challenge lies in performing accurate transcription under these complex interaction patterns, particularly when no prior knowledge about speakers’ number, order, or identity is available. This requires the system to distinguish and organize utterances by the speaker while maintaining the transcription quality expected from modern ASR models.

Early approaches to MS-ASR typically follow a modular pipeline design and can be broadly categorized into two types. The first type relies on speaker diarization to split the

audio into multiple segments based on the predicted target-speaker voice activities [5]–[7]. These segments are then fed into a standard ASR model for transcription. In overlapping regions, additional speech separation techniques [10]–[12] can be applied prior to transcription. The second type of pipeline performs speaker diarization and ASR independently and in parallel [8], [9]. After obtaining transcriptions from the ASR system, time-aligned word boundaries are estimated through the forced alignment technique. These timestamped transcriptions are then matched to diarization outputs via a so-called orchestration process, which attempts to attribute each utterance to a speaker label based on temporal information. However, both paradigms leverage well-established components and suffer from several drawbacks. Errors from one module (e.g., diarization or alignment inaccuracies) can propagate to the final output. Also, the independent components are trained separately and lack a shared optimization objective.

More recent advances include end-to-end frameworks such as Permutation Invariant Training (PIT) [13]–[15] and Serialized Output Training (SOT) [16]–[19]. However, due to the complexity of PIT, the performance of PIT-based systems tends to degrade as the number of speakers increases. In contrast, SOT-style models avoid limiting the maximum number of speakers. Nevertheless, since related content from different speakers is concatenated, and the grammatical structure of utterances in meeting scenarios is often suboptimal, these models require strong long-context awareness and cross-utterance modeling. Furthermore, they lack temporal alignment capabilities, which prevents them from producing time-stamped transcriptions. This type of work, which involves predicting only speaker labels and transcriptions without considering timing information, is usually referred to as Speaker-Attributed ASR (SA-ASR) [19].

Semi-End-to-End (Semi-E2E) approaches, such as Target-Speaker ASR (TS-ASR) [20]–[24], have emerged as a promising alternative. These models accept speaker embedding as input and generate transcriptions for that target speaker, allowing the prediction of the results for each speaker individually. However, most TS-ASR systems face two limitations: (1) the transcribed text cannot be temporally aligned with speaker diarization outputs; (2) they often operate on one speaker at a time, missing contextual information in conversations.

In this work, we propose a new semi-end-to-end MS-ASR framework that addresses the limitations of TS-ASR by

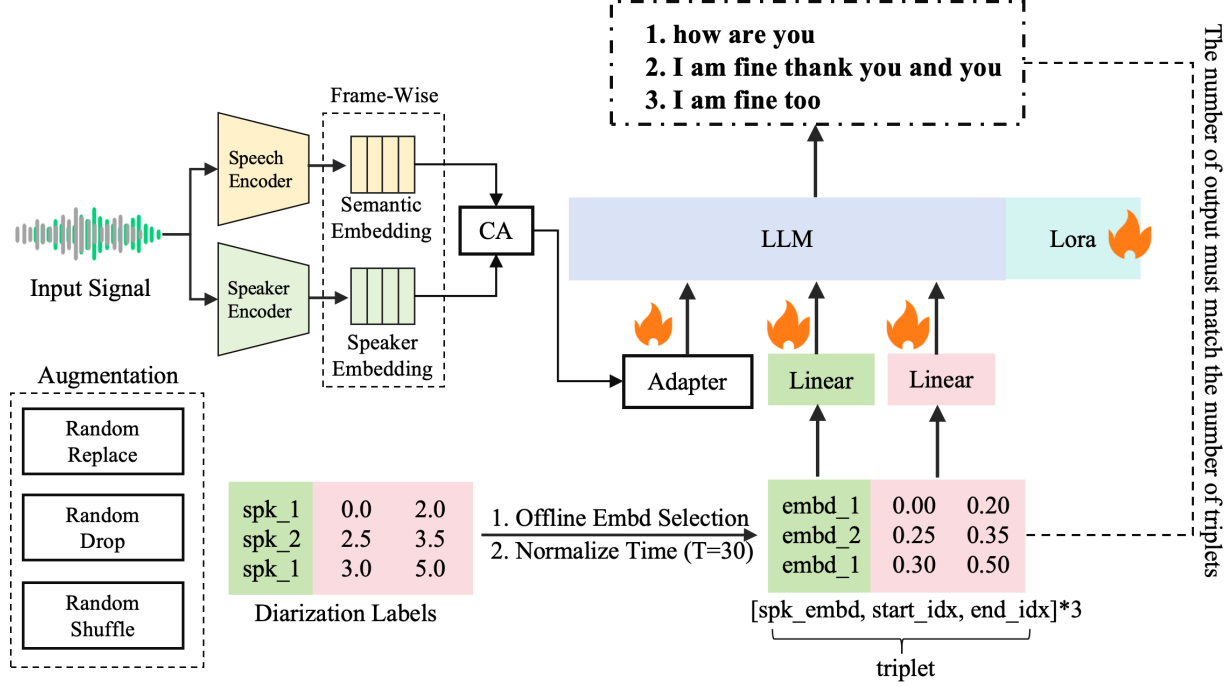


Fig. 1. An overview of our framework.

introducing a triplet-based enrollment mechanism, where each enrolled target speaker is represented by a tuple consisting of (i) a speaker embedding, (ii) a start time of sentence, and (iii) an end time of sentence. This representation allows the model to jointly transcribe speech and output utterance-level timestamps that are directly aligned with a diarization system. Furthermore, we extend the MS-ASR architecture by integrating the large language models (LLMs) [25]–[27], enabling the system to simultaneously process multiple enrolled speaker triplets. This design facilitates contextual modeling across speakers and utterances, improving the efficiency and accuracy of transcription in highly overlapped multi-speaker scenarios.

We evaluate our system in the AliMeeting dataset [28], a real-world Mandarin meeting corpus, and further test its generalization to the MLC-SLM Challenge¹ - Task 2. We adopt tcpWER [29] as our primary evaluation metric, which jointly measures transcription accuracy and temporal alignment. Experimental results demonstrate the effectiveness of our method in producing high-quality, speaker-attributed transcriptions in challenging conversational settings.

II. METHOD

A. Model Backbone

As illustrated in Fig. 1, our framework integrates a large language model (LLM) with two parallel frame-wise encoders:

- A speech encoder that extracts semantic embeddings $\mathbf{H}^s = [\mathbf{h}_1^s, \mathbf{h}_2^s, \dots, \mathbf{h}_T^s] \in \mathbb{R}^{T \times d_s}$ from the input audio,

where T denotes the number of frames and d_s the semantic embedding dimension.

- A speaker encoder that produces speaker-discriminative features $\mathbf{H}^p = [\mathbf{h}_1^p, \mathbf{h}_2^p, \dots, \mathbf{h}_T^p] \in \mathbb{R}^{T \times d_p}$, where d_p denotes the speaker embedding dimension.

To enable speaker-adaptive semantic modeling, we propose a gated cross-attention mechanism that dynamically integrates speaker information with semantic features. The process consists of two key steps: First, semantic features $\mathbf{H}^s$ (from the speech encoder) attend to speaker features $\mathbf{H}^p$ through cross-attention:

$$\mathbf{H}^{ca} = \text{Cross-Attention}(\mathbf{Q} = \mathbf{H}^s, \mathbf{K} = \mathbf{H}^p, \mathbf{V} = \mathbf{H}^p) \quad (1)$$

where $\mathbf{H}^s$ serves as queries to selectively aggregate relevant speaker characteristics from $\mathbf{H}^p$. Second, the attended features are then adaptively gated and combined with the original semantic features:

$$\mathbf{H}^o = \sigma(\mathbf{W}_g \mathbf{H}^{ca}) \odot \mathbf{H}^{ca} + \mathbf{H}^s \quad (2)$$

This gating mechanism, controlled by the sigmoid function $\sigma(\cdot)$, allows flexible modulation of speaker influences while the residual connection preserves essential semantic information. Finally, refined features $\mathbf{H}^o$ are projected through adapter layers to align with the input space of the LLM.

B. Diarization-Aware Triplet Enrollment

As shown in Fig. 2, the whole inputs can be roughly divided into three parts: instructions, multi-modal inputs and labels. The instructions provide the objective and constraints of our task, and the labels are used for generation. As for the multi-modal inputs, to explicitly incorporate diarization awareness

¹<https://www.nexdata.ai/competition/mlc-slm>

into the LLM, we construct a structured triplet representation comprising (1) the target speaker embedding obtained through offline embedding selection and (2) normalized start and end times computed as frame index/total frames for the audio chunk, where both temporal boundaries undergo identical normalization. As illustrated in Fig. 1, this triplet formulation captures essential diarization elements - speaker identity and precise temporal boundaries - in a unified representation that facilitates LLM integration.

These triplets describe speaker identity and utterance-level time boundaries are linearly projected into the LLM input space. The triplet inputs are fed as conditioning instructions, guiding the model in decoding the speech content corresponding to the given speaker and time interval. Unlike traditional TS-ASR methods that rely solely on speaker embeddings or time ranges, our approach combines both, enabling finer speaker-utterance disambiguation, especially under heavily overlapped conditions. Moreover, since the LLM can process multiple triplets simultaneously, our system supports joint decoding of multiple utterances from multiple speakers, with contextual modeling across speaker turns.

C. Data Augmentation Strategies

To improve robustness against diarization errors, we propose three data augmentation techniques during training.

1) *Embedding Replacement*: During training, we randomly replace the original speaker embeddings in some triplets with other speaker embeddings (probability: $p_{replace}$), while setting their corresponding transcription labels to empty. This simulates the imperfect speaker embedding extraction in real-world scenarios and forces the model to ignore incorrect speaker information.

2) *Embedding Dropout*: During training, we randomly drop entire triplets (probability: p_{drop}) and their associated labels. This simulates the presence of missing diarization outputs and forces the model to strictly follow the available diarization input rather than attempting to transcribe all speech content, ensuring output alignment with given triplet sequences.

3) *Triplet Shuffling*: During training, we randomly shuffle the input triplets (probability: $p_{shuffle}$) while simultaneously reordering their corresponding labels. This guarantees that the model aligns outputs with the given enrollment order, regardless of actual chronological order.

D. Chunk-Based Inference

To handle long-form recordings, we implement chunk-based inference. The system first splits prolonged speaker segments into fixed-duration chunks (default: 30s), then organizes them into coherent processing units while enforcing constraints on maximum chunk duration, speaker segment counts per chunk, and total segments per chunk. Cross-chunk continuity is preserved by maintaining consistent speaker embeddings and temporal alignment. This design ensures scalable processing of extended conversations while retaining accurate diarization-to-transcription mapping.

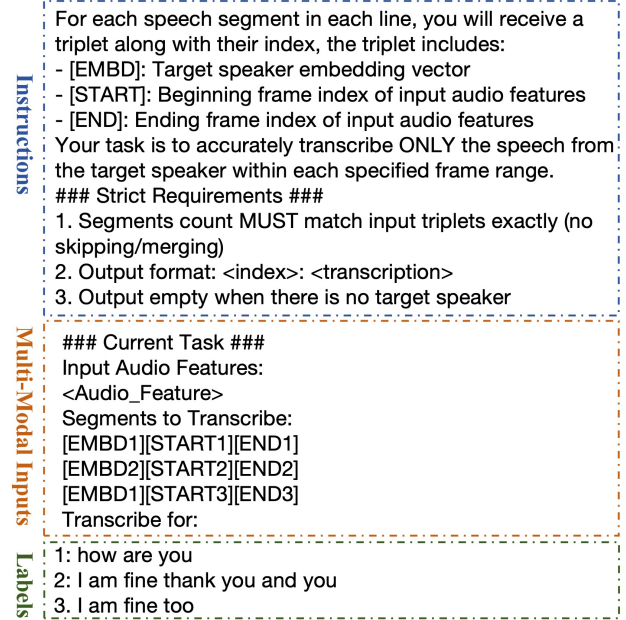


Fig. 2. The construction of our inputs (three sentences from two speakers).

III. EXPERIMENTAL SETTINGS

A. Dataset Usage

To support the pretraining of large-scale neural networks for multi-speaker ASR, we construct a synthetic dataset using two publicly available corpora: Common Voice [30] and VoxBlink2 [31], [32]. Common Voice is a crowd-sourced multilingual ASR dataset; we use only its English subset, which provides high-quality transcribed utterances suitable for supervised ASR training. VoxBlink2 is a large-scale speaker recognition dataset collected from YouTube. Although it lacks transcriptions, it offers substantial speaker diversity and is suitable for generating speaker embeddings. To enable supervised learning, all VoxBlink2 utterances are transcribed using Whisper-Large-V3 model [33] as the ground truth, which finally obtains approximately 4 million high-quality utterance-text pairs, covering over 40,000 unique speakers. During training, we simulate multi-speaker conversational segments by randomly sampling and mixing utterances from different speakers. This mixing is performed on-the-fly to control the number of speakers within each mixture. This simulation strategy allows the model to pretrain on rich and diverse multi-speaker inputs with explicit speaker attribution, providing a strong foundation before fine-tuning on real-world conversational datasets.

To fine-tune our model on real-world conversational data, we adopt the AliMeeting dataset, a publicly available Mandarin meeting corpus for highly-overlapped multi-talker scenarios. The corpus contains approximately 120 hours of recorded meetings involving 2 to 4 speakers per session, covering diverse room sizes and acoustic conditions. We use only the single-channel recordings from the far-field 8-channel microphone array to match our intended deployment setting. Each session includes transcriptions with speaker and